SAGE Module

Explainable Feature Guidance

AST Feature Extraction

28 structural code features

XGBoost Regressor

Features → Fitness (AOCC)

TreeSHAP Analysis

Most influential feature per parent

Natural Language Guidance

"Increase cyclomatic complexity"

DE-Only Constraints

- ✓ rand/1, best/1, current-to-best/1
- ✓ Binomial / Exponential crossover
- ✗ No PSO, CMA-ES, gradients
- ✗ No scipy / finite-differences

Parent code

Inject guidance

Evolutionary Search Loop

1. Initialise Population

Random DE algorithm variants

2. Tournament Selection

From MAP-Elites Archive (size = 3)

3. Build Mutation Prompt

DE constraints + SHAP feature guidance

4. Open-Source LLM

Generate / Mutate DE Algorithm Code

5. Evaluate on MA-BBOB

10 training instances · AOCC metric

6. Compute 4-D Behavioral Descriptor

Adaptation · Recovery · Strategy · Complexity

7. Update MAP-Elites Archive

4 × 4 × 4 × 4 = 256 cells

Next Generation

Selected parent (from archive)

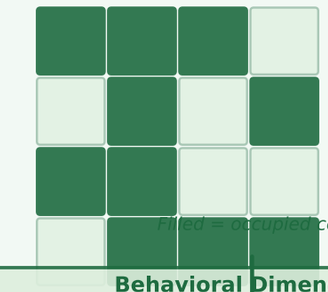
Fitness + descriptor

QD Module

Quality-Diversity Archive

MAP-Elites Archive

4-D space → 4×4×4×4 = 256 cells



Filled = occupied cell

Behavioral Dimensions

- **D1: Adaptation** — Adaptive F, CR, population control
- **D2: Recovery** — Stagnation detection & restart
- **D3: Strategy** — rand/1, best/1, current-to-best/1
- **D4: Complexity** — Code length & cyclomatic depth

Best Candidate per Method

Selected for validation

Validation (100 held-out instances)

5-D · 10-D · 20-D | AOCC

Search Loop | SAGE — Explainability | QD — MAP-Elites Archive | Open-Source LLM | Evaluation / Output